

Qn	Suggested MS
1	
(a)(i)	$\frac{\Delta v}{v} = \frac{1}{2} \left(\frac{\Delta T}{T} \right) + \frac{1}{2} \left(\frac{\Delta l}{l} \right) + \frac{1}{2} \left(\frac{\Delta m}{m} \right)$ $= \frac{1}{2} \left[\frac{0.1}{1.8} + \frac{2}{126} + \frac{0.2}{5.1} \right]$
	$= 0.055$ $= 5.5\%$
	OR
	Find max and min value
	Take average and correct answer
(a)(ii)	$v = \sqrt{\frac{(1.8)(1.26)}{(5.1 \times 10^{-3})}}$ $= 21.08805 \text{ m s}^{-1}$
	$\frac{\Delta v}{v} = 0.055$ $\Delta v = 0.055(21.00805)$ $= 1 \text{ m s}^{-1}$ $v = (21 \pm 1) \text{ m s}^{-1}$
(b)(i)	Average value of 20.7 m s^{-1} falls within the range due to experimental error.
(b)(ii)	Random error
	Repeat the experiment more times so that the average value will tend towards the true value.

Anglo-Chinese Junior College
H2 (9749) Physics

2021 H2 Preliminary Exam Paper 2 MS